

Silicon Photonics MZI Design – Week 3

UBCx Phot1x – Silicon Photonics | Aayushi | edX Username: aayushi_757

1. Design Objective

The objective of this design is to implement an unbalanced Mach-Zehnder Interferometer (MZI) using the SiEPIC EBeam PDK. The circuit consists of an input grating coupler, a 1×2 Y-branch splitter, two waveguide arms with different path lengths, a 2×1 Y-branch combiner, and an output grating coupler. The path-length difference produces wavelength-dependent optical interference.

2. MZI Layout

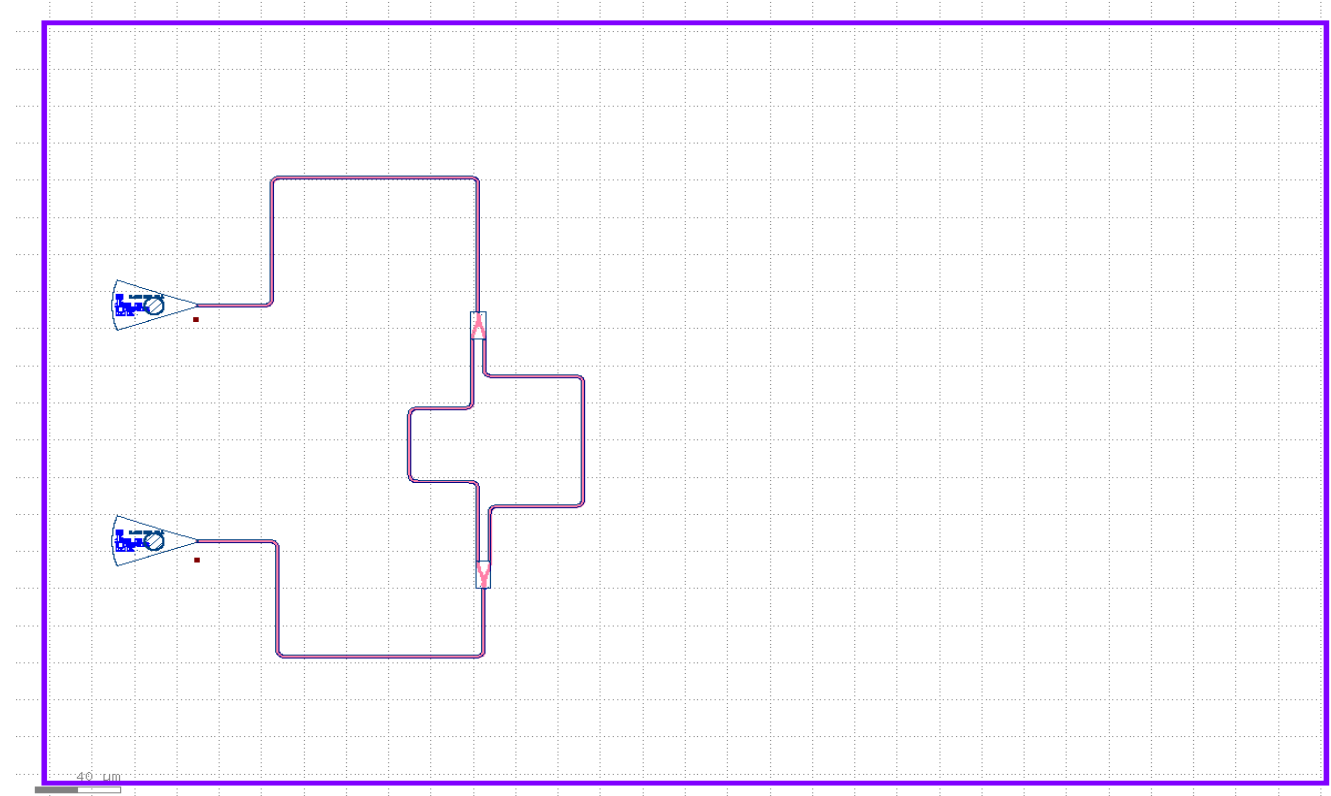


Figure 1. Final MZI layout in KLayout.

3. Main Design Parameters

Parameter	Value
Waveguide width	500 nm
Short arm length	181.61 μm
Long arm length	210.65 μm
Path-length difference	29.04 μm
Grating couplers	TE 1550 nm
Splitter / combiner	EBeam Y-branches
PDK	SiEPIC EBeam v0.4.53

4. Operating Principle

Light entering through the input grating coupler is divided by the first Y-branch into the two waveguide arms. Because the arms have different lengths, the two optical fields accumulate different phases. At the second Y-branch, the fields interfere constructively or destructively depending on wavelength, producing a periodic wavelength-dependent transmission response.

5. Simulated Spectral Response

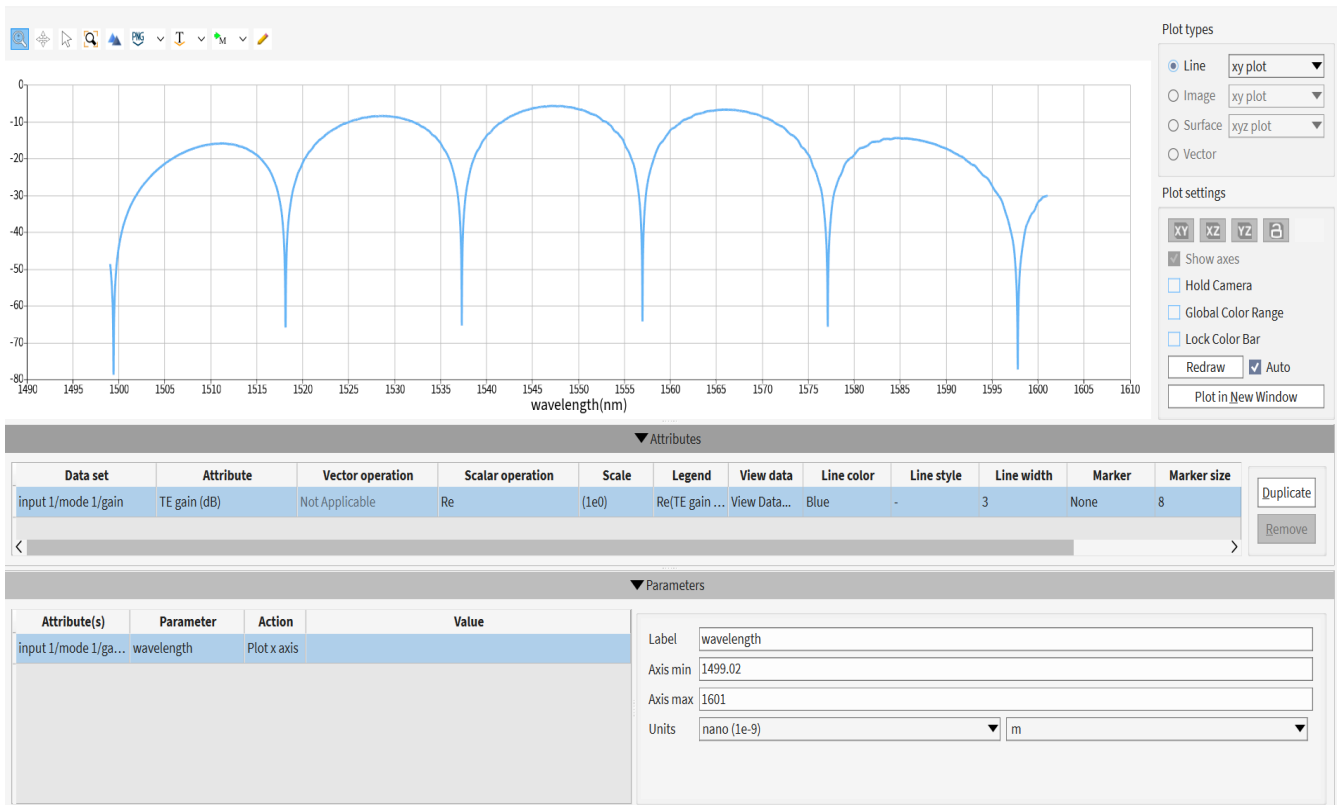


Figure 2. Simulated TE gain response over approximately 1500–1600 nm. The periodic maxima and minima are characteristic of wavelength-dependent interference in an unbalanced MZI.

6. Layout Verification and Submission

The final GDS layout was checked using the SiEPIC verification tools and submitted through the openEBL GitHub workflow. The pull request completed with 7 successful automated checks and was successfully merged into the main course repository.

Submission item	Value
GDS filename	EBeam_aayushi_757.gds
GitHub fork	aayushi19-ece/openEBL-2026-10
Submission folder	submissions/
Automated verification	7 checks passed
Repository status	Merged into SiEPIC/openEBL-2026-10 main

7. Summary

The design implements an unbalanced MZI with a 29.04 μm arm-length difference and a 500 nm-wide waveguide. Its wavelength-dependent interference response was simulated, and the submitted GDS successfully passed the automated GitHub verification workflow.